17.7.4.1 Step 1: Add PHP 8.4 Repository and Install Dependencies

We will use the **ondrej/php** PPA to install PHP 8.4 and its extensions.

```
sudo apt update && sudo apt upgrade -y
sudo add-apt-repository ppa:ondrej/php -y
sudo apt update
sudo apt install nginx php8.4 php8.4-fpm php8.4-mbstring php8.4-xml php8.4-bcmath php8.4-cu
```

This installs Nginx, PHP 8.4, and the necessary PHP extensions for Laravel. After installation, check the PHP version to confirm:

```
php -v
```

17.7.4.2 Step 2: Install Composer

Composer is a dependency manager for PHP, essential for Laravel. Install it globally:

```
cd ~
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer
```

Make sure Composer is installed correctly:

```
composer --version
```

You should see the Composer version output.

17.7.4.3 Step 3: Clone Laravel 12 Application

We will clone a sample Laravel 12 application from GitHub. Replace `your-repo` with your actual repository.

```
cd /var/www
sudo git clone https://github.com/your-repo/laravel12-app.git laravel12
cd laravel12
sudo chown -R www-data:www-data .
```

> *If your git repository is private, you may need to set up SSH keys or use HTTPS with credentials.*